SquashVAR: A Single-Camera Geometry-Aware Assistant Referee for Squash In/Out Line Calls

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Abstract

Squash line calls remain difficult to automate in an accessible, reproducible way: while at least one commercial single-camera product exists, public details, evaluation protocols, and end-to-end implementations are largely unavailable, limiting transparency and research progress. We present SquashVAR, an open, single-camera geometry-aware assistant-referee pipeline for real match footage. The system detects and tracks the ball under motion blur, occlusion, and glass-court reflections, assigns observations to court surfaces, maps image points into metric court coordinates using planar homographies, and identifies wall/floor impacts from trajectory-based motion signatures. We evaluate ball detection on a fixed 3,000-frame benchmark and assess end-to-end impact detection against frame-level ground truth under both surface-aware and surface-agnostic matching. In/out calls are derived from court geometry using rule-based thresholds with an abstention [21] band near boundaries. We release the evaluation protocol and derived artifacts to support reproducibility and future extensions.

1. Introduction

Squash line calls are high-stakes and time-critical, yet they are still largely judged by a single broadcast-style view. Automated squash officiating remains comparatively underexplored [12, 13, 20] in the research literature, despite the sport's distinct visual constraints: a very small and fast-moving ball, frequent player occlusions, and strong glare/reflections from glass courts. [4] These factors reduce the reliability of standard ball-tracking pipelines when applied to real match footage.

At the same time, practical deployment is constrained: multi-camera installations are expensive and uncommon outside elite venues, so a single-camera approach is the most realistic pathway to a widely usable assistant-referee tool. Our goal is therefore not to introduce a new detector architecture, but to build a complete, geometry-aware single-camera system that can translate noisy 2D detections into court-referenced events that are meaningful for officiating (impacts and in/out decisions), and to report an end-to-end evaluation with a reproducible workflow.

We present SquashVAR, a single-camera, geometry-aware assistant-referee pipeline. Given a match video, SquashVAR (i) detects the ball, (ii) links detections into tracks, (iii) assigns each detection to a court surface region, (iv) maps image points into metric court coordinates using per-surface planar homographies, and (v) detects wall/floor impacts from trajectory signatures. A final decision layer applies rule-based court geometry (tin/out-lines) with an abstention band near boundaries to avoid over-confident calls.

Novelty claim. The novelty of SquashVAR is the end-to-end integration of detection, tracking, court-surface assignment, homography-based metric grounding, and trajectory-based impact extraction into a single reproducible single-camera pipeline for squash officiating analysis, rather than any single component in isolation.

Contributions. We provide:

- An end-to-end single-camera assistant-referee pipeline that produces court-referenced impact events and supports rule-based in/out decisions.
- A fixed 3,000-frame benchmark definition for detector reporting, and a frame-level impact ground truth set (N=78) for end-to-end impact evaluation.
- An evaluation of impact detection under surface-aware and surface-agnostic matching with a clear tolerance protocol.
- A reproducible implementation with scripts, artifact formats, and commands to regenerate detector reports, world-coordinate mapping, and impact outputs.

2. Related Work

2.1. Squash-specific vision research

Automated squash officiating is still relatively sparse compared to adjacent sports, and existing work typically addresses only parts of the full problem. Prior squash-focused studies examine referee decision-making and learning-based decision support in constrained settings [12,13], while single-camera pipelines have been explored for drill-oriented analysis with more controlled viewpoints and assumptions [20]. These efforts motivate the feasibility of squash automation, but they leave open several practical gaps that matter for match officiating.

In particular, the literature provides limited end-to-end systems that (i) operate on broadcast-style match video with severe occlusions and glass-court reflections, (ii) explicitly ground detections into metric court coordinates via calibration and surface reasoning, and (iii) report frame-level impact-event evaluation with a clear tolerance protocol. Our work targets these gaps by presenting a single-camera, geometry-aware pipeline that converts 2D detections into court-referenced trajectories and impact events, enabling rule-based in/out reasoning with an abstention band near boundaries.

2.2. Transferable methods from adjacent sports

Tiny-object tracking in sports motivates temporal context and robust tracking. TrackNet proposes a heatmap formulation for high-speed ball tracking in tennis [6]; temporal feature multiplexing has been applied to table tennis ball tracking [8]. Broadcast hockey work explores puck localization under occlusion [2]. For court grounding, homographies and calibration are standard tools in monocular sports analytics [5, 22].

2.3. Real-time detection and tracking

We do not propose a new detector; rather, we adopt a modern YOLO-family model as a practical backbone for tiny-object localization under strict latency constraints. [7] In sports video, tracking-by-detection pipelines commonly pair a fast detector with an association mechanism to maintain identities through brief dropouts and occlusions. [1] SquashVAR follows this paradigm: we use YOLOv8 for ball detection [7] and BoT-SORT for robust association into tracklets under intermittent visibility [1]. Unlike approaches that operate purely in image space, our downstream stages exploit geometry: detections are assigned to court planes and mapped to metric coordinates via planar homographies, enabling physically meaningful event rules and court-line reasoning [5, 22]

3. Data

3.1. Benchmark video

Benchmark video. We evaluate on a fixed 3,000-frame benchmark (1920×1080, 60 fps) extracted from *Farag v Elias, Paris Squash 2024 Semi-Final* [17](YouTube id: yMKDa4aHMsk; accessed 2025-10-17). We release frame indices and our annotations, not the source video.

3.2. Training dataset

Training dataset. We curated `ball_ft_v2` (3,001 images, 6,277 labels) and split it ∼80/10/10 (train/val/test) using a fixed, reproducible split definition (file lists stored with the dataset to avoid randomness across runs). Evaluation uses the separate fixed 3,000-frame benchmark video.

3.3. Impact ground truth

Annotation task. We manually annotated N=78 ball–surface impacts on the fixed 3,000-frame benchmark at frame-level resolution. Each annotation records `frame_idx` and a surface label in {FRONT, FLOOR, LEFT, RIGHT}. The resulting counts are: FRONT 34, FLOOR 28, LEFT 11, RIGHT 5. The ground-truth file is released as `data/gt/impacts_gt.csv`.

3.4. Licensing

Licensing. We share only derived frame indices, detector labels, calibration files, and our impact annotations. Redistribution of the YouTube source video is not included.

4. Methods

4.1. Pipeline overview

Pipeline: video → detections → tracks → plane assignment → world coordinates → impacts → in/out.

We standardize surface labels throughout as FRONT, LEFT, RIGHT, FLOOR.

Detector. We use the *Ultralytics* implementation of YOLOv8 [7], specifically the `yolov8s` architecture, fine-tuned for a single class (ball). All training and inference are performed with the Ultralytics `yolo` CLI/`ultralytics` package (version: `v8.2.103`), using the final checkpoint `/workspace/runs/detect/train8/weights/best.pt`. We report benchmark results at two inference sizes (`imgsz`=1920 and 1280).

Tracker. Tracks are formed with BoT-SORT [1] through the Ultralytics tracking pipeline [7], using the stock configuration file `configs/trackers/botsort.yaml` (ReID disabled). For tracking we use `classes=0`, `max_det=5`, `conf=0.45`, `iou=0.50`, and `half=True`.

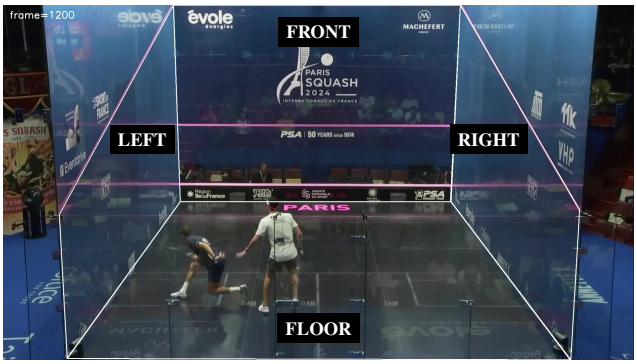


Figure 1. **Court calibration overlay for plane assignment.** Labeled polygonal regions (FRONT, FLOOR, LEFT, RIGHT) gate plane selection: each detection is assigned to a plane (with margins) before homography mapping to court coordinates.

Table 1. Impact-detection thresholds (cm / frames unless noted).

Proximity	$D_{\text{FRONT}} = D_{\text{BACK}} = D_{\text{SIDE}} = 20$ cm
Acceleration	$A_{\text{FRONT}} = 120,\ A_{\text{WALL}} = 120$
Speed-drop	$R_{\text{FRONT}} = 0.6,\ R_{\text{WALL}} = 0.6$
Heading change	$\Theta_{\text{FRONT}} = 40^\circ,\ \Theta_{\text{WALL}} = 40^\circ$
Debounce	`GAP_MIN` $= 6$ frames
Floor segments	len 1–8, conf ≥ 0.35, down/up ≥ 2 px, debounce 6

Training setup. We fine-tune with AdamW [11] (or SGD [18]), cosine LR scheduling [10], and EMA of weights [16]. Detections use non-maximum suppression (NMS) [15].

4.2. Court geometry: homographies and plane assignment

Each frame uses four planar homographies—front wall, floor, left wall, right wall—built from four-point pixel↔world correspondences [5]. We test multiple candidate points per bounding box (center and three lower-edge variants), compute signed distance to each plane polygon, and select the plane with maximum positive distance (within per-plane margins). The chosen pixel point is mapped to court coordinates (x, y, z) in centimeters where x increases from front to back, y increases from left to right, and z increases from floor to ceiling.

4.3. Trajectory + calculus impact detection

Tracks are smoothed with a 5-frame moving window (FPS=60). We compute velocity, acceleration magnitude, and heading change. Wall impacts require proximity to the nearest surface and motion signatures; all thresholds are in Table 1. Corner-seam hits near FRONT×LEFT or FRONT×RIGHT emit two events.

4.4. In/Out logic

In/out is rule-based on court geometry: front out-line height 457 cm, side/back out-line height 213 cm, tin height 48 cm [14]. We abstain within ±2 cm of any decision boundary (`NO_DECISION`); otherwise we label `IN` or `OUT`. In/out is applied after impact detection at the surface-appropriate coordinate.

Table 2. Detection performance on the fixed 3,000-frame benchmark. Best values per column are in bold (latency: lower is better).

Model	ImgSz	P	R	mAP@50	mAP@50–95	ms/img
YOLOv8-S	1920	**0.937**	0.816	0.906	0.487	22.4
YOLOv8-S	1280	0.862	**0.890**	**0.911**	0.461	11.6
YOLOv8-n	1280	0.835	0.622	0.711	**0.544**	**8.0**

Table 3. Impact detection on the 3,000-frame benchmark (primary reporting: ±25 frames).

Mode	TP	FP	FN	P	R	F1
surface-aware	41	10	37	0.804	0.526	0.636
surface-agnostic	47	4	31	0.922	0.603	0.729

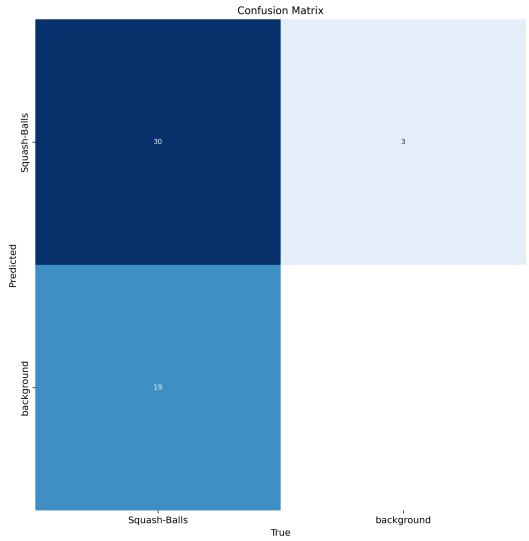


Figure 2. **Confusion matrix** on the 3,000-frame benchmark.

5. Experiments

5.1. Detection results

Metrics. We report Ultralytics Val metrics: precision (P) and recall (R) are computed at the confidence threshold that maximizes F_1 [19]; mAP@50 is mean Average Precision at IoU=0.50; and mAP@50–95 is COCO-style mAP averaged over IoU thresholds 0.50:0.95. [9, 19] The reported ms/img is the average per-image inference time (as logged by Ultralytics) for the given setting.

Results and takeaway. Table 2 compares accuracy and speed across three settings. YOLOv8-S at 1920 achieves the highest precision, while YOLOv8-S at 1280 improves recall and mAP@50 with substantially lower latency. The YOLOv8-n 1280 baseline is fastest, but its recall and mAP@50 drop noticeably relative to YOLOv8-S, reflecting the accuracy–latency trade-off for tiny-object detection in this footage.

5.2. End-to-end impact detection

We evaluate impact detection against the frame-level ground truth described in Section 3.3. Matching is performed on `frame_idx` with a ± 25-frame tolerance for primary reporting. We report both *surface-aware* matching (surface must agree) and *surface-agnostic* matching (surface ignored).

Impact evaluation metrics. We compute true positives (TP), false positives (FP), and false negatives (FN) by matching predicted impacts to ground-truth impacts within a ± 25-frame tolerance. Precision $P = \frac{\text{TP}}{\text{TP+FP}}$, recall $R = \frac{\text{TP}}{\text{TP+FN}}$, and $F_1 = \frac{2PR}{P+R}$. Surface-aware matching additionally requires the predicted surface label to match the ground-truth surface; surface-agnostic matching ignores surface labels.

5.3. In/Out qualitative

Decision rule. In/out is computed deterministically from the mapped impact location and the relevant rule boundary for the contacted surface (Sec. 4.4). For each predicted impact, we evaluate the corresponding world coordinate (cm) against the line for that surface (e.g., tin and front out-line on FRONT; side/back out-lines on LEFT/RIGHT; out-line on FLOOR if applicable). To avoid over-confident calls near boundaries, we return `NO_DECISION` when the impact lies within ± 2 cm of any decision boundary; otherwise we output `IN` or `OUT`.

Qualitative verification protocol. We performed a qualitative sanity check by sampling impacts across different surfaces and reviewing the overlay video that visualizes (i) the detected impact frame, (ii) the assigned surface, and (iii) the mapped court coordinate used by the decision rule. This check verifies that the in/out logic is applied consistently with the calibrated geometry and that the abstention band activates near lines. We do not claim a quantitative in/out accuracy because we do not yet have a dedicated frame-level in/out ground-truth dataset; building such a dataset is a priority for future work.

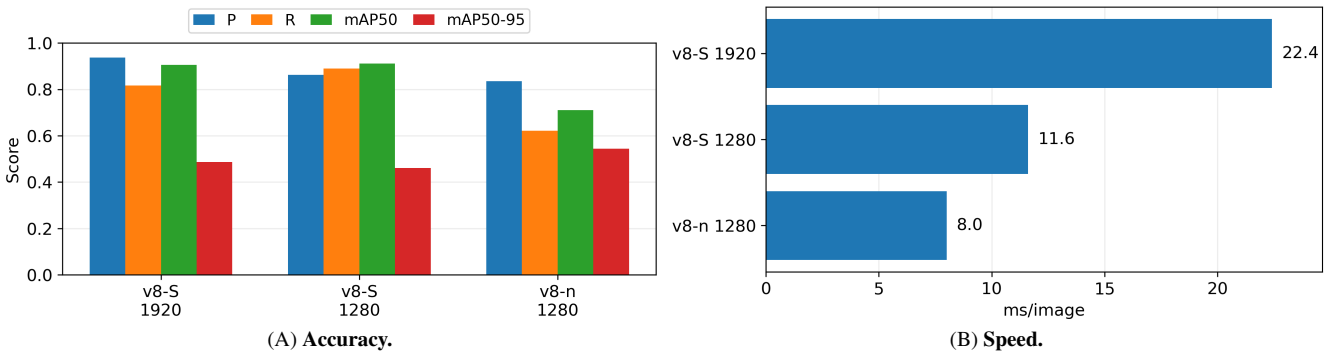
6. Discussion

Performance varies strongly by surface. Front-wall impacts are detected reliably, while floor events remain challenging under occlusion and reflections; the segment-based floor rule improves recall when calculus signatures are weak. Right-wall performance is poorest (F1=0 on this benchmark, with only $N = 5$ right-wall ground-truth impacts), which is consistent with two practical issues we observed: (i) few clear right-wall contacts appear in the training/validation footage, limiting exposure to that impact signature, and (ii) the right-wall region occupies a narrow image area in the broadcast view, making plane assignment and proximity gating more brittle. Because N is small, the right-wall metric is also highly sensitive to a handful of misses. Future work will expand labeled right-wall impact coverage, retune surface-specific thresholds/margins for RIGHT, add a second annotator pass to compute inter-rater reliability (κ), and construct a dedicated in/out evaluation set with explicit line-call ground truth (rather than qualitative spot-checking).

7. Reproducibility

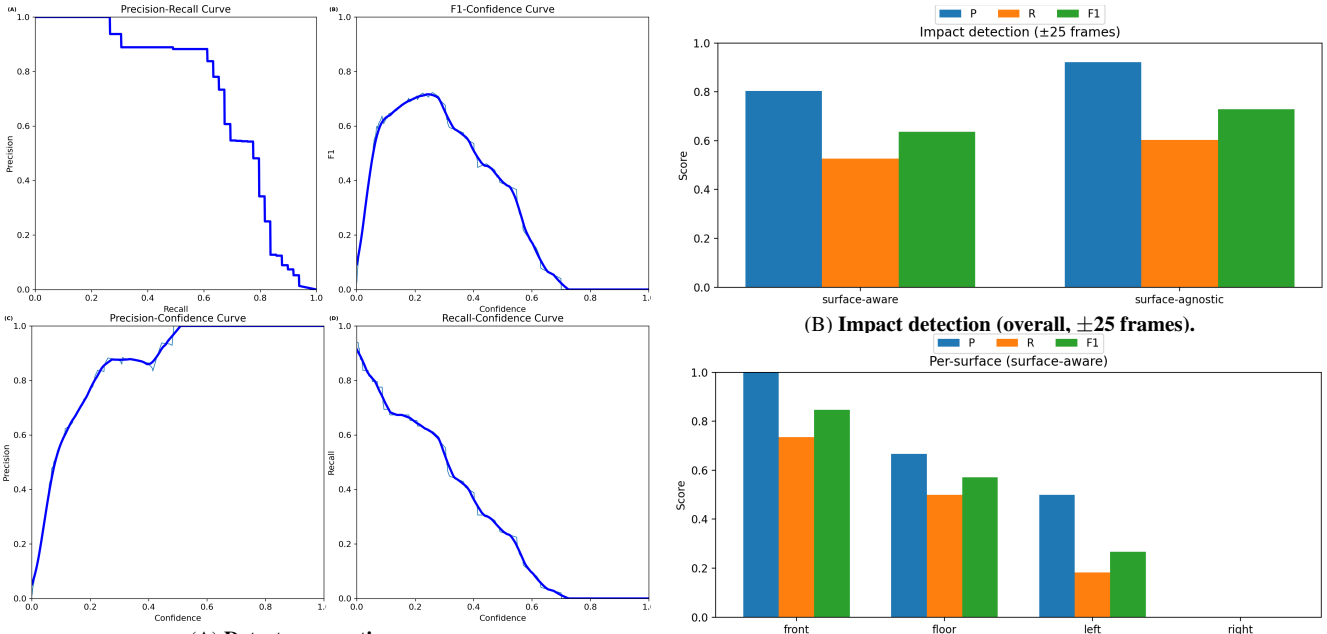
Code and artifacts: `https://github.com/mahmoud-yassin10/SquashVAR` [21]

We provide a deterministic, script-based pipeline and release all non-copyrighted artifacts needed to reproduce our reported results (detector outputs, calibration YAMLs, and impact ground truth). End-to-end reproduction follows four steps: (i) detect+track the ball with Ultralytics YOLOv8 and BoT-SORT, (ii) map tracked detections to court coordinates using polygon-gated plane assignment (implemented with OpenCV polygon tests) [3] and fixed per-plane homographies [5, 22] (`tools/world_from_labels_modes.py` with `configs/calib.*.yaml`), (iii) extract impacts with fixed smoothing and threshold rules (`tools/nano_impacts.py`), and (iv) evaluate against our frame-level annotations (`data/gt/impacts_gt.csv`) using tolerance-based matching (`tools/eval_impacts_pr.py`). A complete run script with exact commands is included in the repository documentation.



(A) **Accuracy.** (B) **Speed.**

Figure 3. **Detection performance on the 3,000-frame benchmark.**



(A) **Detector operating curves.** (B) **Impact detection (overall, ±25 frames).** (C) **Impact detection (per-surface, surface-aware).**

Figure 4. **Detector diagnostics and end-to-end impact detection.**

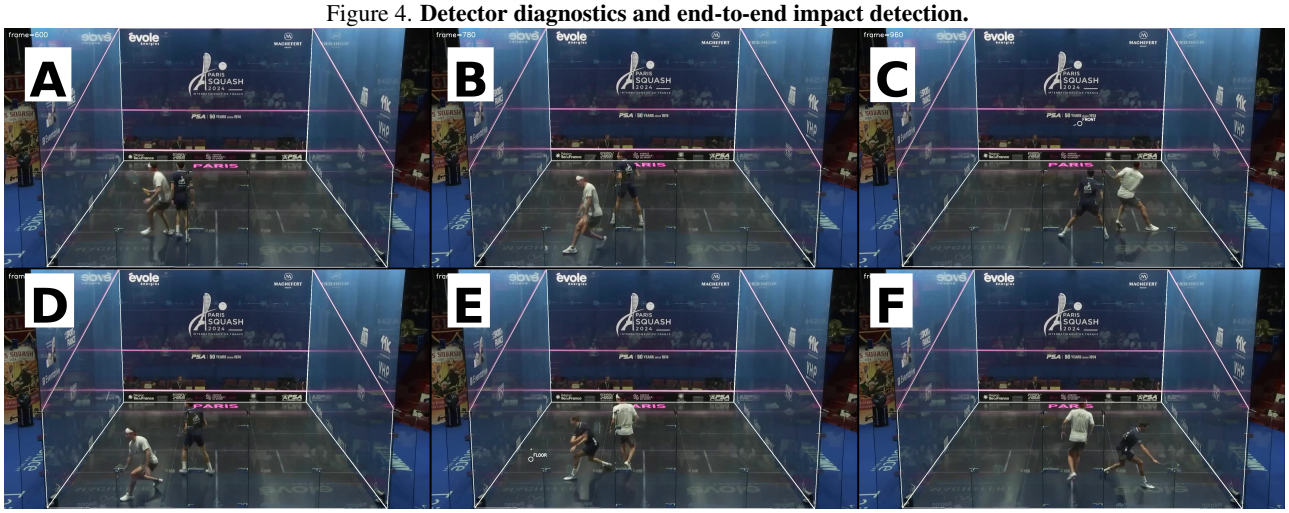


Figure 5. Qualitative impact outcomes. (**C,E**) detected impacts; (**A,B,D,F**) missed impacts, often due to occlusion, glare/reflections, or seam ambiguity in single-camera glass-court footage.

8. Conclusion

We presented SquashVAR, a single-camera, geometry-aware assistant-referee pipeline for squash that addresses the sport's underexplored officiating setting where the ball is tiny, fast, occluded, and glass courts introduce reflections. SquashVAR solves this by combining (i) ball detection and tracking, (ii) plane-aware assignment and homography mapping into court coordinates, and (iii) trajectory-calculus rules to localize wall and floor impacts used for downstream line calls. On a fixed 3,000-frame benchmark, YOLOv8-S achieves strong ball detection at two operating points (1920: P=0.937, R=0.816, mAP@50=0.906; 1280: P=0.862, R=0.890, mAP@50=0.911). On end-to-end impact detection with frame-level ground truth (N=78) and a $\pm$25-frame tolerance, SquashVAR attains F1=0.636 under surface-aware matching and F1=0.729 when surface is ignored. Finally, in/out decisions are computed from mapped impact coordinates using explicit rule thresholds with an abstention band near boundaries to avoid over-confident calls. Limitations include sparse right-wall events in the benchmark and single-camera ambiguity near seams; future work will expand annotations and evaluation to stronger in/out protocols.

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